

SIMATS ENGINEERING ASSESSMENT TOOL 1

COURSE CODE: CSA0706

COURSE NAME: COMPUTER NETWORKS

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. (BL3)

Assessment Tool Weightage: 40%

Annexure A

ANALYTICAL PROBLEM SOLVING

Code of Conduct:

I _____V.Lasya(192524186)_____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: v.lasya

Annexure A

ANALYTICAL PROBLEM SOLVING

1. Suppose we want to transmit the message 11100011 and protect it from errors using the CRC polynomial X^3+1 .

- i.) Use polynomial long division to determine the message that should be transmitted.
- ii) Suppose the leftmost bit of the message is inverted due to noise on the transmission link. What is the result of the receiver's CRC calculation? How does the receiver know that an error has occurred?

1. Cyclic Redundancy Check (CRC)

Given:

Data = 11100011

Generator Polynomial = X^3+1

Generator bits = 1001

Degree of polynomial = 3

i) Determine the transmitted message

Step 1: Append three zeros

Since the degree of the polynomial is 3, append 3 zeros.

Original data:

11100011

After appending zeros:

11100011000

Step 2: Perform Modulo-2 (XOR) Division

Generator = 1001

Dividend = 11100011000

$1110 \oplus 1001 = 0111$

Bring down next bit

$1110 \oplus 1001 = 0111$

Bring down next bit

$1111 \oplus 1001 = 0110$

Bring down next bit

$1100 \oplus 1001 = 0101$

Bring down next bit

$$1011 \oplus 1001 = 0010$$

Bring down remaining bits

Final remainder = 100

CRC bits = 100

Step 3: Transmitted Message

Original Data : 11100011

CRC : 100

Transmitted Message = 11100011100

Answer (i):

Transmitted message = 11100011100

ii) Leftmost bit is inverted

Received message becomes

01100011100

Now divide by the same generator (1001).

The remainder obtained is

Remainder = 001

Since

remainder $\neq$ 000

the CRC check fails.

How does the receiver detect the error?

Receiver divides the received message by the generator polynomial.

If the remainder is 000, the message is accepted.

If the remainder is non-zero, an error has occurred.

Answer (ii):

Received message = 01100011100

CRC remainder = 001 (non-zero)

Therefore, the receiver detects an error because the remainder is not zero.

2. a) A network with bandwidth of 10 Mbps can pass only an average of 12,000 frames per minute with each frame carrying an average of 10,000 bits. What is the throughput of this network?

b) A network has a bandwidth of 3000 Hz (300 to 3300 Hz) assigned for data communications. The signal- to-noise ratio is 3162. Find the channel capacity

2(a). Throughput Calculation

Given

Bandwidth = 10 Mbps

Frames/minute = 12,000

Frame size = 10,000 bits

Formula

Throughput=

Time

Frames×Frame Size

Calculation

Bits transmitted per minute

$12000 \times 10000 = 120000000$ bits

Convert to bits/sec

60

120000000

=2000000 bps

=2 Mbps

Answer

Throughput = 2 Mbps

2(b). Channel Capacity

Given

Bandwidth

$B = 3000$ Hz

Signal-to-noise ratio

$$S/N=3162$$

Shannon Capacity Formula

$$C = B \log_2$$

$$2(1+S/N)$$

Substitute values

$$C = 3000 \log_2$$

$$2(3163)$$

Since

$$\log_2(3163) \approx 11.63$$

Therefore

$$C = 3000 \times 11.63$$

$$C \approx 34890 \text{ bps}$$

$$C \approx 34.9 \text{ kbps}$$

Answer

Channel Capacity ≈ 34.9 kbps

3. A multispecialty hospital uses wireless patient-monitoring devices to continuously transmit patients' heart rate and blood pressure readings to a central monitoring station. Since the data is transmitted over a wireless channel, electromagnetic interference occasionally changes one bit during transmission. To ensure that doctors receive accurate patient information without waiting for retransmission, the hospital implements the **Hamming (12,8)** error-correcting code.

3. Hamming (12,8) Error-Correcting Code

Given

Data bits = 8

Code word = 12 bits

Parity bits = 4

Error = One bit changes during transmission

Working

The Hamming (12,8) code places parity bits at positions

1, 2, 4, 8

Data bits occupy

3,5,6,7,9,10,11,12

The sender calculates the parity bits and transmits the 12-bit codeword.

During wireless transmission, suppose one bit changes because of electromagnetic interference.

The receiver recalculates the parity bits and forms the syndrome.

Syndrome = 0000 → No error

Syndrome ≠ 0000 → Indicates the exact position of the erroneous bit

The receiver flips that bit back to its correct value.

Advantages

Detects single-bit errors.

Corrects single-bit errors automatically.

No retransmission required.

Doctors receive accurate patient-monitoring data immediately.

Improves reliability in wireless communication.

Answer

The Hamming (12,8) code automatically detects and corrects the single-bit error by using four parity bits. The receiver computes the syndrome, identifies the incorrect bit position, corrects it, and recovers the original patient data without requesting retransmission.

4. a) A company uses the IP address **172.16.10.0/24** and has implemented subnetting using a **/27 subnet mask** for its internal departments. Using the given subnet mask, apply subnetting techniques to calculate the total number of subnets created and the number of usable hosts per subnet. Determine the subnet to which the IP address **172.16.10.78** belongs. Identify the network address and broadcast address for that subnet. Also, determine whether the IP address **172.16.10.95** falls within the same subnet range based on the given configuration.

b) A network uses the Internet Protocol version 6 address **2001:0db8:0000:0000:ff00:0042:8329**. Apply IPv6 addressing rules to compress the given address into its shortest form and then expand the compressed address back to its full notation. Using a prefix length of **/64**, determine the network prefix of the given address. Also, calculate the total number of possible host addresses available within this network based on the given prefix length.

4(a). IPv4 Subnetting

Given

Network

172.16.10.0/24

Subnet mask

/27

Step 1: Number of Subnets

Borrowed bits

$27 - 24 = 3$

Number of subnets

$2^3 = 8$

Total subnets = 8

Step 2: Hosts per Subnet

Remaining host bits

$32 - 27 = 5$

Total hosts

$2^5 = 32$

Usable hosts

$32 - 2 = 30$

Usable hosts per subnet = 30

Step 3: Subnet Ranges

	Subnet Network	Broadcast
1	172.16.10.0	172.16.10.31
2	172.16.10.32	172.16.10.63
3	172.16.10.64	172.16.10.95
4	172.16.10.96	172.16.10.127
5	172.16.10.128	172.16.10.159
6	172.16.10.160	172.16.10.191
7	172.16.10.192	172.16.10.223
8	172.16.10.224	172.16.10.255

Step 4: IP Address 172.16.10.78

78 lies between

64 – 95

Therefore

Network Address = 172.16.10.64

Broadcast Address = 172.16.10.95

Step 5: IP Address 172.16.10.95

95 is the broadcast address of the same subnet.

Hence it belongs to the same subnet but cannot be assigned to a host.

Answer

Total subnets = 8

Usable hosts/subnet = 30

172.16.10.78 belongs to subnet 172.16.10.64/27

Network address = 172.16.10.64

Broadcast address = 172.16.10.95

172.16.10.95 is the broadcast address of the same subnet.

4(b). IPv6 Addressing

Given

2001:0db8:0000:0000:0000:ff00:0042:8329

Compressed Form

2001:db8::ff00:42:8329

Expanded Form

2001:0db8:0000:0000:0000:ff00:0042:8329

Network Prefix (/64)

First 64 bits

2001:0db8:0000:0000::/64

Host Addresses

Remaining bits

$128 - 64 = 64$

Possible host addresses

$$2^{64}=18,446,744,073,709,551,616$$

Approximately

$$1.84 \times 10$$

19

Answer

Compressed address = 2001:db8::ff00:42:8329

Expanded address = 2001:0db8:0000:0000:0000:ff00:0042:8329

Network prefix = 2001:0db8:0000:0000::/64

Number of host addresses = $2^{64} = 18,446,744,073,709,551,616$

5. A university is conducting an online semester examination for **8,000 students** simultaneously. At exactly **10:00 AM**, the examination server begins transmitting encrypted question papers to all students. The university server is connected through a **10 Gbps high-speed network**, while students access the examination portal using different internet connections such as **fiber broadband (100 Mbps), home Wi-Fi (20 Mbps), and mobile networks (5–10 Mbps)**. During the first few minutes of the examination, many students with slower internet connections report incomplete downloads, repeated retransmissions, delayed acknowledgments, and temporary freezing of the examination portal. Network administrators observe that the server continues transmitting data at a high rate even though several student devices are unable to receive and process packets at the same speed. This results in receiver buffer overflow, unnecessary retransmissions, increased network traffic, and delays in delivering the question papers. **Analyze** the above scenario and identify the primary networking problem. Explain how the mismatch between the sender's transmission speed and the receiver's processing capability affects communication.

5. Network Analysis

Primary Networking Problem

The main networking problem is Flow Control.

Explanation

The university server sends encrypted question papers at a very high speed (10 Gbps), while many students use much slower internet connections (5–100 Mbps). As a result, several student devices cannot receive and process packets as quickly as they are transmitted.

This mismatch causes the following problems:

Receiver buffers become full (buffer overflow).

Packets are dropped before they can be processed.

Lost packets must be retransmitted.

Acknowledgements are delayed.

Network traffic increases due to repeated retransmissions.

Students experience incomplete downloads, freezing, and delays in receiving the question papers. Flow control mechanisms ensure that the sender adjusts its transmission rate according to the receiver's processing capability, preventing buffer overflow and improving reliable communication.

Answer

The primary networking problem is flow control. The sender transmits data faster than the receivers can process it, leading to receiver buffer overflow, packet loss, retransmissions, delayed acknowledgements, increased network congestion, and slower delivery of the examination question papers. Flow control prevents these issues by regulating the data transmission rate to match the receiver's capacity.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate	Selects correct	Inappropriate or partially	Unable to select

		method/formula with strong justification	method with minor errors	correct method	method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation